# Intermediate Elective 2

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## Set Up

### Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)

# text stuff
library(ggtext)
```

### Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality
water_quality <- read_xlsx(here("data", "Aquatic Sampling
 ↪  Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

## Cleaning

```
# creating new object from sites
ncos_sites <- sites %>%
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```
# creating new object from taxon_list
taxon_list_clean <- taxon_list %>%
  # clean taxon names
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```
# creating new clean object from water quality
water_quality_clean <- water_quality %>%
  # clean column names
  clean_names() %>%
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") %>%
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) %>%
  # group by date_site column
  group_by(date_site) %>%
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
```

```
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) %>%
  # ungroup data frame
  ungroup() %>%
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

```
# creating new object from aquatic inverts
aq_ins_clean <- aq_ins %>%
  # clean column names
  clean_names() %>%
  # filtering join to only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) %>%
  # renaming columns
  rename(date = date_on_vial) %>%
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) %>%
  # filter to only include filter beaker samples
  filter(sample_type %in% c("FB 250", "FB250")) %>%
  # convert to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") %>%
  # group by date, site, taxon
  group_by(date, site, taxon) %>%
  # calculate average abundance per liter
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) %>%
  # ungroup after
  ungroup() %>%
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) %>%
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) %>%
  # join with taxon list
```

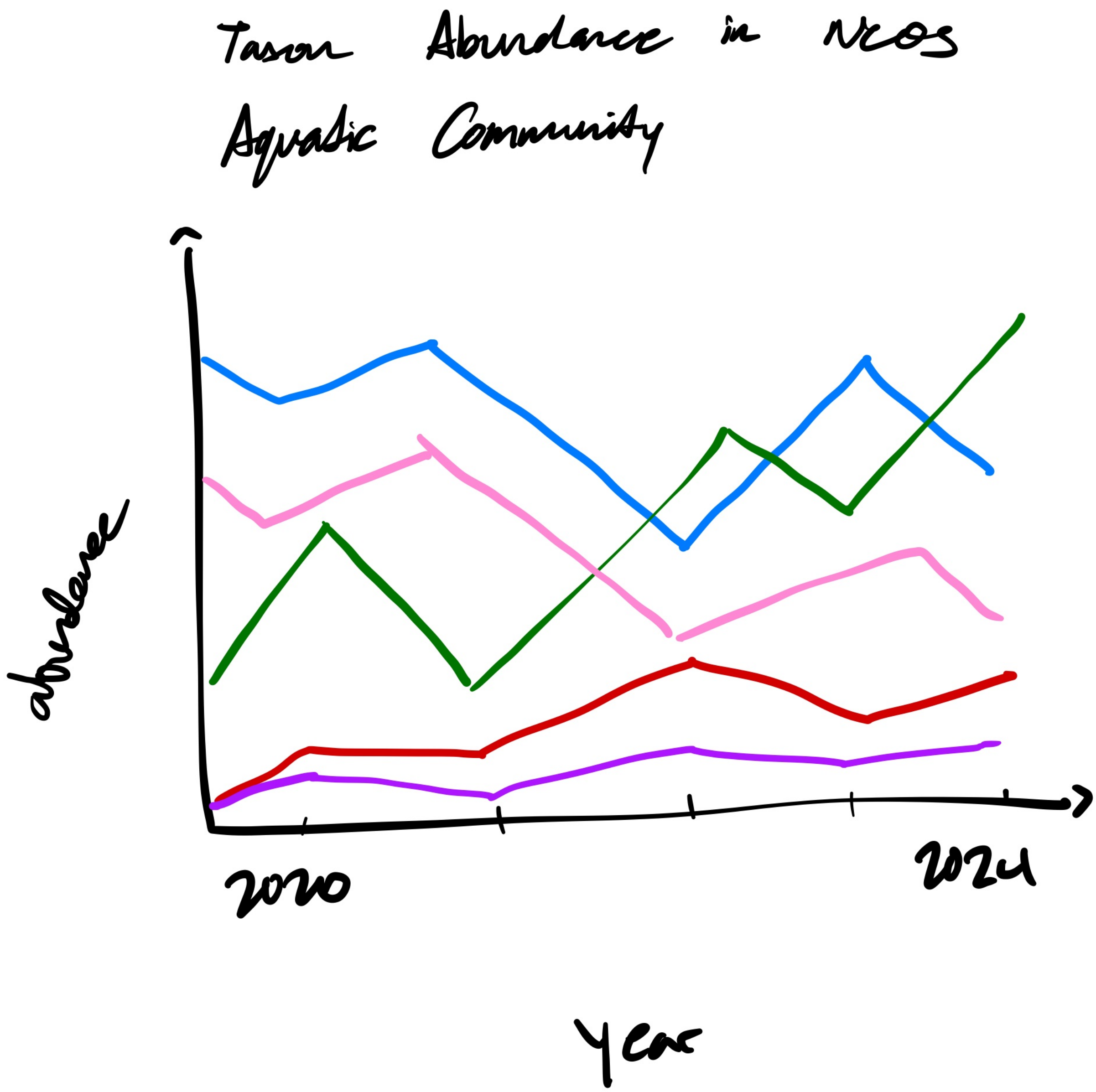
```r
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) %>%
  # full name of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) %>%
  # factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
   ↪ = TRUE),
         site_full = fct_rev(site_full))
```

## Problems

### 1. Explain your inspiration

I chose Steven Ponce's visual of China's electricity generation mix. I thought using this as inspiration made sense since I could do something similar and visualizing a mix of species (or something similar) in the NCOS space. I would probably keep `year` as my x-axis since I am showing my y-variable over time. I would have my `taxon` on the y-axis, and each different one would be represented by a different color.

**2. Plan your figure**



## 3. Code your figure

### Time series

```
taxa_year <- aq_ins_clean %>%
  mutate(year = year(date)) %>%
  group_by(year, taxon) %>%
  summarize(total_ave_lit = sum(ave_lit, na.rm = TRUE), .groups = "drop") %>%
  # including these taxon
  filter(taxon %in% c(
    "ostracod",
    "copepod",
    "hemiptera_corixidae_boatman",
    "cladocera",
    "nematode",
    "diptera_ceratopogonidae",
    "diptera_chironomid",
    "annelida_oligochaete"
  )) |>
  mutate(
    taxon_label = case_when(
      taxon == "ostracod"                    ~ "Ostracod",
      taxon == "copepod"                     ~ "Copepod",
      taxon == "hemiptera_corixidae_boatman" ~ "Corixidae",
      taxon == "cladocera"                   ~ "Cladocera",
      taxon == "nematode"                    ~ "Nematode",
      taxon == "diptera_ceratopogonidae"     ~ "Ceratopogonidae",
      taxon == "diptera_chironomid"          ~ "Chironomid",
      taxon == "annelida_oligochaete"        ~ "Oligochaete"
    ),
    # log + 1 transform to match the copepod abundance transform used in the
    ↪  rest of the analysis
    log_total = log(total_ave_lit + 1)
  )
```

### Color & linetype palettes

```
#} label: visuals

# corresponding colors
```

```r
pal <- c(
  "Copepod"          = "#4fc3f7",
  "Ostracod"         = "#81c784",
  "Cladocera"        = "#ffb74d",
  "Corixidae"        = "#ce93d8",
  "Chironomid"       = "#f48fb1",
  "Ceratopogonidae" = "#80cbc4",
  "Nematode"         = "#bcaaa4",
  "Oligochaete"      = "#a5d6a7"
)


#corresponding lines
lty_pal <- c(
  "Copepod"          = "solid",
  "Ostracod"         = "dashed",
  "Cladocera"        = "dotted",
  "Corixidae"        = "longdash",
  "Chironomid"       = "twodash",
  "Ceratopogonidae" = "dashed",
  "Nematode"         = "dotted",
  "Oligochaete"      = "longdash")
```

### Labels

```r
end_labels <- taxa_year %>%
  group_by(taxon_label) %>%
  slice_max(year, n = 1) %>%
  ungroup()
```

### Theme colors

```r
navy_bg  <- "#0a1628"
panel_bg <- "#0d1f38"
grid_col <- "#1e3a52"
text_col <- "#e8e4d8"
sub_col  <- "#8aabbf"
axis_col <- "#6a9ab5"
```

### Making Visual

```
p <- taxa_year %>%
  ggplot(aes(x = year,
             y = log_total,
             color = taxon_label,
             group = taxon_label)) +
  # Lines + points
  geom_line(linewidth = 0.9, na.rm = TRUE) +
  geom_point(size = 2.5, na.rm = TRUE) +

  # Right-side taxon labels (replaces legend)
  geom_text(
    data     = end_labels,
    aes(label = taxon_label),
    hjust    = -0.12,
    size     = 3,
    na.rm    = TRUE
  ) +

  # X-axis: one tick per year, extra right margin for labels
  scale_x_continuous(
    breaks = 2020:2024,
    expand = expansion(mult = c(0.03, 0.22))
  ) +

  # Y-axis: log + 1 scale
  scale_y_continuous(
    name   = "log(avg. individuals / L  +  1)",
    breaks = 0:7,
    expand = expansion(mult = c(0.05, 0.1))
  ) +

  # Palettes (no legend - labels carry that information)
  scale_color_manual(values = pal,      guide = "none") +
  scale_linetype_manual(values = lty_pal, guide = "none") +

  # Titles
  labs(
    title    = "Copepods and ostracods dominate NCOS invertebrate
     ↪  communities",
    subtitle = paste0(
```

```
    "Sum of average individuals per liter (FB 250 samples, NCOS quarterly
     ↪ sites)  \u00b7  ",
    "log(x\u00a0+\u00a01) transformed  \u00b7  2020\u20132024"
  ),
  caption = paste0(
    "SOURCE: Aquatic Sampling Data 2020\u20132025  \u00b7  ",
    "Sites: NDC, NEC, NMC, NPB, NPB1, NPB2, NVBR  \u00b7  ",
    "Sample type: filtered beaker 250\u00a0mL"
  ),
  x = NULL
) +

# making theme
theme_minimal(base_size = 11) +
theme(
  plot.background  = element_rect(fill = navy_bg,  color = NA),
  panel.background = element_rect(fill = panel_bg, color = NA),
  panel.grid.major = element_line(color = grid_col, linewidth = 0.35),
  panel.grid.minor = element_blank(),
  axis.text        = element_text(color = axis_col, size = 9),
  axis.title.y     = element_text(color = axis_col, size = 9,
                                  margin = margin(r = 8)),
  axis.ticks       = element_line(color = grid_col),
  plot.title       = element_text(color = text_col, size = 15,
                                  face = "bold", margin = margin(b = 4)),
  plot.subtitle    = element_text(color = sub_col, size = 9.5,
                                  margin = margin(b = 14)),
  plot.caption     = element_text(color = "#4a6a80", size = 7.5,
                                  hjust = 0, margin = margin(t = 10)),
  plot.margin      = margin(20, 24, 14, 20)
)
```

### Save

```
ggsave(
  filename = "aquatic_taxa_timeseries.png",
  plot     = p,
  width    = 10,
  height   = 6,
  dpi      = 300,
```

```
  bg       = navy_bg)

message("Saved \u2192 aquatic_taxa_timeseries.png")
```

## 4. Describe your process

I forked his repository in order to use the same code to build this figure. I basically just copied and pasted everything, but I didn't realize that he used custom code to build his figure and it was a little difficult trying to figure out how to get it to run on my end. I thought it was also cool that you could do that. I would say this visual is a lot less boring to look at than figures that were made previously. I think the colors and the cleaner look make the visuals more appealing to look at.